

NANCY ONYEJIKA

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EDUCATION

Northeastern University, Boston, MA
Master of Science in Computer Science
magna cum laude

May 2025

Bachelor of Science in Computer Science and Behavioral Neuroscience, Minor in Biology
Honors Program, cum laude
Dean's List: Fall 2020 – Fall 2023

May 2024

TECHNICAL SKILLS

Languages:	Kotlin Java Python TypeScript JavaScript Svelte Ruby
Infrastructure & Cloud:	AWS (Bedrock, Batch, Lambda, S3, DynamoDB, IAM, SSM, Route53, CloudWatch) Terraform TeamCity Docker
Data & Storage:	PostgreSQL MySQL MongoDB Apache Lucene Redis Flyway Ktorm
Frameworks & Tools:	SvelteKit Node.js http4k Koin Gradle JUnit 5 Git TeamCity Sentry

PROFESSIONAL EXPERIENCE

CarGurus, Cambridge, MA

Algorithmic Traffic Acquisition Engineer II

Jun 2024 – Sept 2023

- Architected and deployed an AI-powered vehicle recommendation engine using Claude (AWS Bedrock) to generate semantic embeddings for 11M+ listings, achieving <115ms P99 latency via Lucene HNSW KNN indexing, S3-backed index bootstrapping, and high-throughput concurrent producer-consumer pipelines.
- Designed and built GRID, a full-stack internal platform (SvelteKit, Kotlin, Terraform) from inception to production, integrating LLM-driven natural language to JSON query generation; reduced recommendation testing time from minutes to seconds and became the primary debugging and validation interface for the recommendation engine.
- Extended and productized the filter/v2 recommendation DSL into a fully configurable system by implementing conditional execution predicates, synthetic personas, fit scoring, and user history filtering, enabling product, personalization, and external teams to define complex recommendation strategies without code changes.
- Founded and led a recurring AI Retrospective program across engineering, establishing a structured framework to evaluate AI tooling adoption, identify friction points, and drive actionable improvements tracked in Jira, accelerating organization-wide AI integration and developer productivity.
- Drove cross-functional engineering initiatives including a codebase-wide null-safety migration, development of AI-assisted developer tooling adopted team-wide, and decommissioning of legacy inventory infrastructure (DynamoDB, TeamCity, Terraform), improving system reliability, maintainability, and operational efficiency.

Algorithmic Traffic Acquisition Engineer I

Jun 2024 – Apr 2023

- Led architecture and rollout of Stardate, a multi-region distributed job orchestration platform with automatic retries, exponential backoff, and observability tooling, replacing legacy cron and Ruby services and eliminating recurring production incidents caused by duplicate job execution.
- Engineered a high-performance persona-to-listing matching engine using bitwise binary encoding (BigInteger) with wildcard masking, reducing inventory update runtime from 24 hours to ~20 minutes across millions of records by replacing iterative attribute comparisons with constant-time bitwise evaluation and Caffeine-cached representations.

CarGurus, Cambridge, MA

Software Engineering Co-op, Algorithmic Traffic Acquisition

Jan 2023 – Sept 2023

- Optimized legacy Hillclimber algorithms for channels bidding on partners, including Google and Facebook, on an aggregated level, driving an 18% surge in client return on investment for ad spending.
- Refactored a Ruby-based spending threshold web page to ReactTS, resulting in improved load times and internal user satisfaction. The migration of both frontend and backend to Kotlin and MySQL boosted maintainability and scalability. The inclusion of new features like raw governor setting display, and direct editing of database values, contributed to a 15% increase in decision-making accuracy.
- Engineered a TypeScript, Kotlin, React, and MySQL solution for comprehensive spending analysis and publication across clients, yielding a 20% reduction in data collection and analysis time. Utilized RESTful APIs to streamline data retrieval and integration with external systems.
- Designed and implemented daily Slack alerts summarizing spending patterns, boosting accessibility, intervention speed, and decision accuracy by 12%. Successfully led the team-wide shift from plain to rich text Slack alerts.
- Pioneered an alert management infrastructure through Kotlin, Ktorm, and SQL migrations, slashing spam from obsolete alerts by 175%, and ensuring only relevant and actionable alerts reach the team.
- Led engineering triage, meticulously monitoring our Sentry portal for production bugs, errors, and exceptions. Ensured swift resolution through direct intervention, collaboration, or delegation, thereby upholding system integrity and performance.

PROJECTS

Master's Project: Automatic Code Refactoring with Large Language Models

[Research Paper](#)

- Engineered and curated a dynamic JavaScript "ground truth" repository for code quality analysis, integrating Gemini API to enable real-time AI-driven code smell detection and refactoring aligned with evolving best practices.
- Contributed to the design of an interactive AI-assisted code review tool, enabling teaching assistants to efficiently assess and deliver feedback on student submissions, improving scalability of code evaluation.

Artificial Intelligence in Poker: A Comparative Analysis of Decision-Making Using Search Algorithms

[Research Paper](#) | [GitHub](#)

- Developed and assessed three AI algorithms in Python (Expectiminimax, Monte Carlo Tree Search, and Deep Q-Learning) for Texas Hold'em poker, which outperformed standard bots and showcased adaptability in stochastic games through heuristic-driven simulations, enhancing strategic play and providing insights for real-world decision-making.